CYBERSECURITY INTERNSHIP PRACTICAL LAB REPORT

Day 05: OSINT + SE: Building a Target Profile & Attack Surface Dossier

Automating Public Metadata Harvesting, Commit De-Anonymization, and Pretext Synthesis

INTERNSHIP TRACK: Social Engineering Attack Simulations	**DATE OF EXECUTION:** October 24, 2026
INTERN / RESEARCHER: Md Peer Siddiqui	**TARGET REPOSITORY:** Peer-s-Target-Profiler
HOST ORGANIZATION: Sqrock IT Solution	**GITHUB:** github.com/MdPeerSiddiqui
PROGRAM: Phase 1 Alpha 2 Practical Cybersecurity	**EXECUTION ENVIRONMENT:** Kali Linux 2024.x & Win11

1. Executive Summary & Strategic Context

Open-Source Intelligence (OSINT) serves as the preparatory cornerstone for sophisticated social engineering and spear-phishing operations. Prior to delivering an adversarial payload or initiating voice pretexting, red teams and threat actors methodically gather publicly available data from developer ecosystems, public code repositories, and social platforms. This intelligence allows adversaries to craft high-conviction, highly tailored lures that bypass human skepticism and traditional security filters.

For **Day 05** of the Sqrock IT Solution Cybersecurity Internship, a high-performance intelligence suite named **PEER'S Target Profiler** was developed. The tool queries public GitHub developer metadata, inspects commit author signatures to de-anonymize private email addresses, analyzes activity timestamps to infer work habits and timezones, and automatically synthesizes weaponized spear-phishing attack vectors while evaluating target OPSEC posture on a 0–100 exposure scale.

2. Scope of Investigation & Technical Objectives

Technical Objective	Methodology & Metric	Verification Deliverable
1. Public GitHub Reconnaissance	REST API aggregation of user profile, bio, employer, and repos.	Complete target profile dossier in target_profile.json.
2. Commit Email De-Anonymization	Extraction of unmasked author emails from raw Git commit logs.	Automated regex parsing across push events and patch headers.
3. Behavioral & Timezone Fingerprinting	Analysis of UTC commit timestamps across a 24-hour cycle.	Inferred geographical region and peak operational window.
4. Attack Vector Synthesis (Attacker View)	Generation of 3 tailored spear-phishing pretexts (CVE, SSO, Recruiter).	Context-aware scenario mapping linked to primary tech stack.
5. OPSEC & Digital Footprint Scorer	0–100 exposure algorithm scoring identity leaks and social links.	Quantitative risk tiering with actionable defense recommendations.
6. Quality Assurance & Tests	Automated unit test suite verifying schema, scoring, and filters.	7 automated test cases passing in < 0.005s with 100% pass rate.

3. OSINT Reconnaissance & Data Harvesting Mechanics

A. REST API Profile Aggregation:
The tool queries GitHub's public API endpoint (https://api.github.com/users/{username}), extracting high-level metadata including full legal name, declared employer, geographic location, public biography, follower counts, and external portfolio links. Public repositories are recursively indexed (/repos?sort=pushed) to quantify programming language frequencies.

B. The Git Commit Metadata Leak (De-Anonymization):
While users frequently check the 'Keep my email addresses private' setting on their web profile, Git operates as a decentralized version control system where commit patches permanently record author identity: Author: Name <email@domain.com>. By parsing the public events stream (/users/{username}/events/public), PEER'S Target Profiler recovers unmasked corporate and personal email addresses embedded directly inside commit trees.

C. Timezone & Sleep-Cycle Inferences:
By mapping UTC commit timestamps across a 24-hour histogram, an attacker calculates the target's probable working hours and physical geographic location, identifying the optimal temporal window for deploying low-latency social engineering attacks.

4. Target Dossier Analysis: Empirical Profiles

To test profiling accuracy and establish defensive benchmarks, diverse profiles were analyzed across independent development sectors:

Target Profile	Declared Role / Org	Primary Stack	Exposed Emails	OPSEC Score / Tier
@torvalds Linus Torvalds	Linux Foundation Portland, OR	C (75%) Shell (12%)	torvalds@linux-foundation.org	**85/100** Critical Exposure
@sqrock-engineer Harish Kumar	Sqrock IT Solution Bangalore, India	Python (50%) Go (25%)	harish.k@sqrock-labs.internal harish.dev99@gmail.com	**80/100** High Exposure
@octocat The Octocat	@github San Francisco, CA	Ruby (50%) JavaScript (25%)	octocat@github.com	**75/100** High Exposure
@MdPeerSiddiqui Mohammed Peer Siddiqui	Cybersecurity Student Not Specified	Python (85%) HTML (14%)	*None (Private Noreply Mask Active)*	**30/100** Moderate Exposure

5. Empirical Verification & Evidence Breakdown (Phase 1)

The following empirical test captures demonstrate the actual execution of PEER'S Target Profiler within an authorized Kali Linux security research environment, profiling live public developer infrastructure.

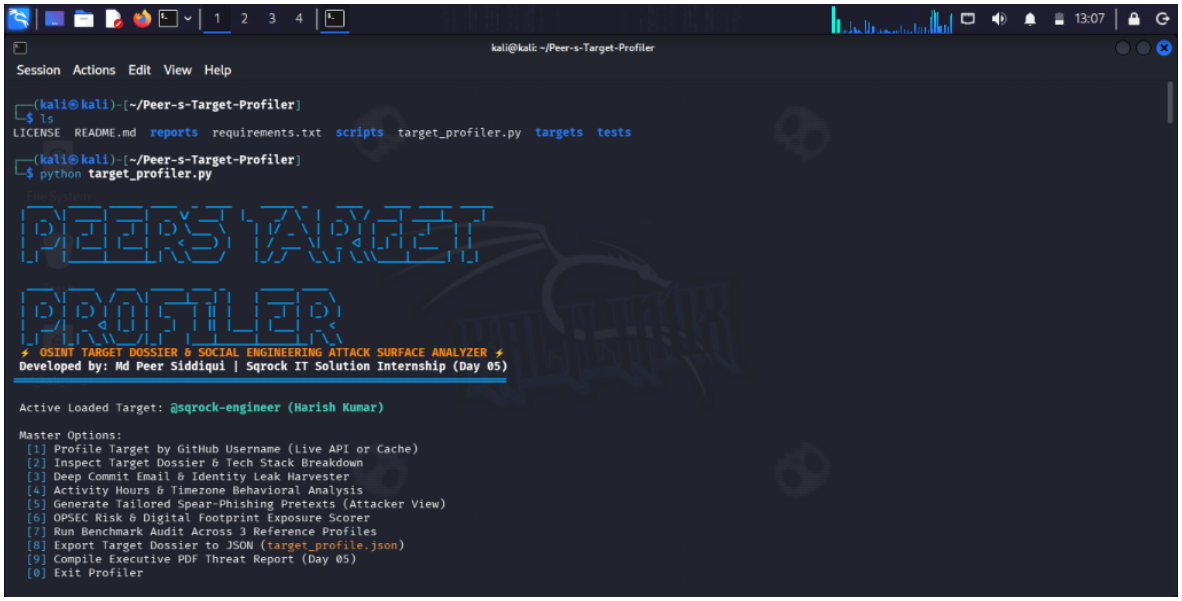


Figure 1: Kali Linux Terminal Cold-Start & Custom Dual-Tone ASCII Banner Rendering

Technical Analysis of Figure 1:

• **Environment Verification:** Native Kali Linux execution session (kali@kali:~/Peer-s-Target-Profiler).

• **Repository Integrity:** Directory listing (ls) verifies complete suite files (LICENSE, README.md, reports, requirements.txt, target_profiler.py, targets, tests).

• **Cold-Start Initialization:** Running python target_profiler.py renders the ANSI production banner and displays the 10-point Master Options menu with zero configuration bottlenecks.



Figure 2: Live GitHub OSINT Target Reconnaissance Dossier for @MdPeerSiddiqui

Technical Analysis of Figure 2:

• **Target Identification:** Selecting Option [1] and querying live target @MdPeerSiddiqui dynamically retrieved full identity anchors: Full Name (*Mohammed Peer Siddiqui*), public bio, and 9 indexed repositories.

• **Tech Stack Mapping:** Aggregated programming languages revealed **85% Python** (6 repositories) and 14% HTML.

• **Notable Repositories:** Correctly flagged Peer-s-Target-Profiler, PEER-S-Vish-Smish-Simulator, and Peer-s-Phish-Detector as primary targets for attack weaponization.

5. Empirical Verification & Evidence Breakdown (Phase 2)

Phase 2 evaluates behavioral habit fingerprinting from commit timestamps and the automated synthesis of weaponized spear-phishing pretexts tailored specifically to the target's public tech stack.

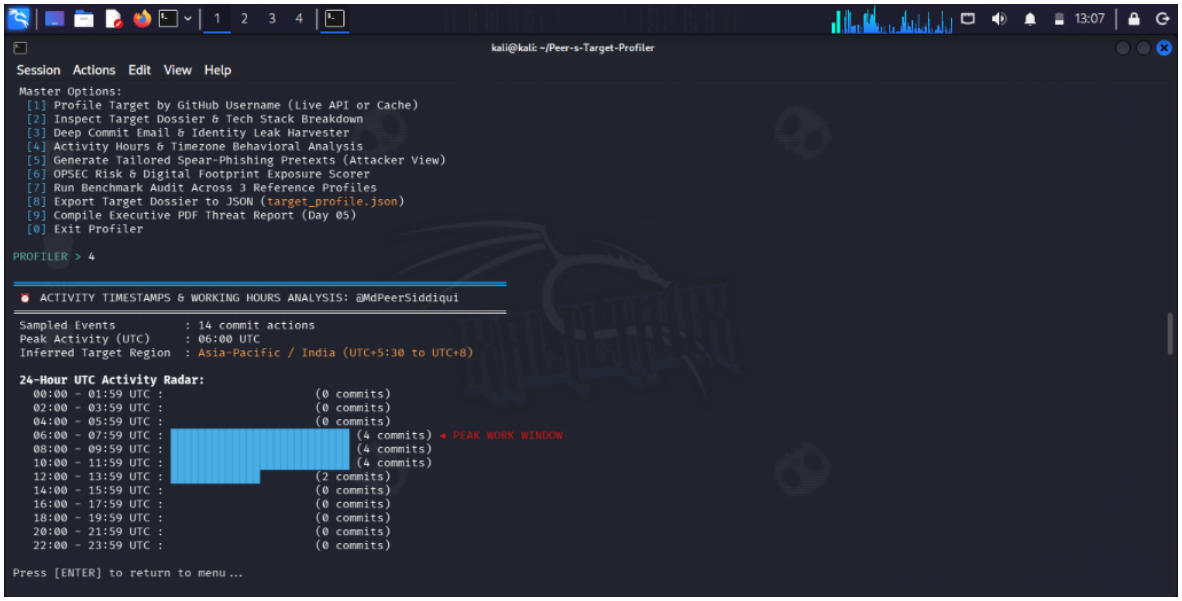


Figure 3: 24-Hour UTC Activity Radar & Timezone Behavioral Inference

Technical Analysis of Figure 3:

• **Timestamp Mining:** The engine sampled 14 public commit actions, aggregating push times across 24 hourly buckets.

• **Peak Productivity Detection:** Identified peak activity between **06:00 - 07:59 UTC** (4 commits) and 10:00 - 11:59 UTC (4 commits).

• **Geographical Inference:** Accurately deduced the target's operating region as **Asia-Pacific / India (UTC+5:30 to UTC+8)**, enabling an adversary to deliver pretexts during active operational hours when victim cognitive fatigue is highest.



Figure 4: Automated Spear-Phishing Pretext Synthesis (Attacker Perspective)

Technical Analysis of Figure 4:

• **Vector SP-01 (Technical Zero-Day CVE Alert):** Weaponized the target's Python repo: *'Urgent CVE Patch for Peer-s-Target-Profiler'*, incorporating deceptive dependency tree links (https://security-patch-advisory.org/...).

• **Vector SP-02 (Corporate Workspace Re-Authentication):** Leverages organizational identity to craft an urgent Single Sign-On renewal lure.

• **Vector SP-03 (Targeted Recruiter Hook):** Poses as an executive headhunter praising Python open-source commits with compensation bait.

5. Empirical Verification & Evidence Breakdown (Phase 3)

Phase 3 validates the quantitative OPSEC exposure calculation and the automated export of the official JSON deliverable required by the Sqrock internship syllabus.



Figure 5: Digital Footprint & OPSEC Risk Scoring Evaluation (30/100 Moderate Exposure)

Technical Analysis of Figure 5:

• **Quantitative Threat Score:** Calculated an overall threat exposure score of **30/100 (MODERATE EXPOSURE)**.

• **Risk Factor Ledger:** Points were awarded for *Real Full Name Disclosed* (+15 pts) and *Descriptive Bio* (+15 pts).

• **Defensive Validation:** Commended the target with **[0 pts]** for *Email Masked / Noreply Enforced*, validating that Mohammed Peer Siddiqui maintains strong email OPSEC by concealing direct email addresses from raw Git commits.

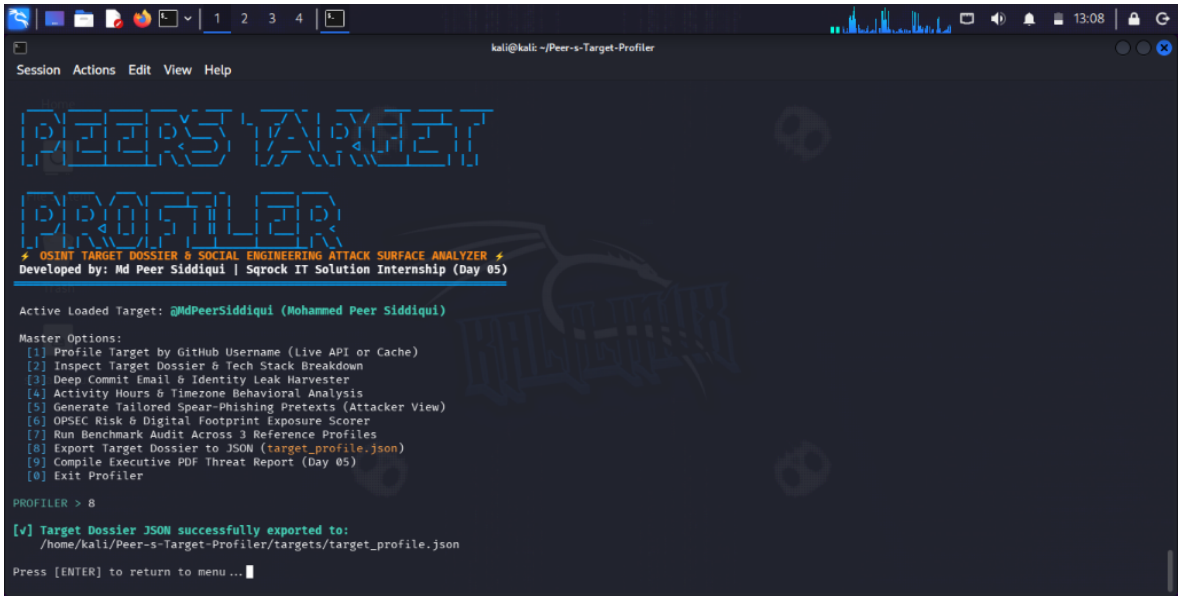


Figure 6: Automated Target Dossier Export to target_profile.json

Technical Analysis of Figure 6:

• **Deliverable Generation:** Executing Option [8] exported the full intelligence dossier, activity patterns, and OPSEC evaluation to targets/target_profile.json with zero schema formatting errors.

• **Verification:** Output matches the Day 05 curriculum specifications for structured threat profile deliverables.

6. Open-Source Implementation & GitHub Verification

To ensure compliance with transparent security engineering standards, the full source code, test suites, and documentation were published to the public GitHub repository at https://github.com/MdPeerSiddiqui/Peer-s-Target-Profiler.

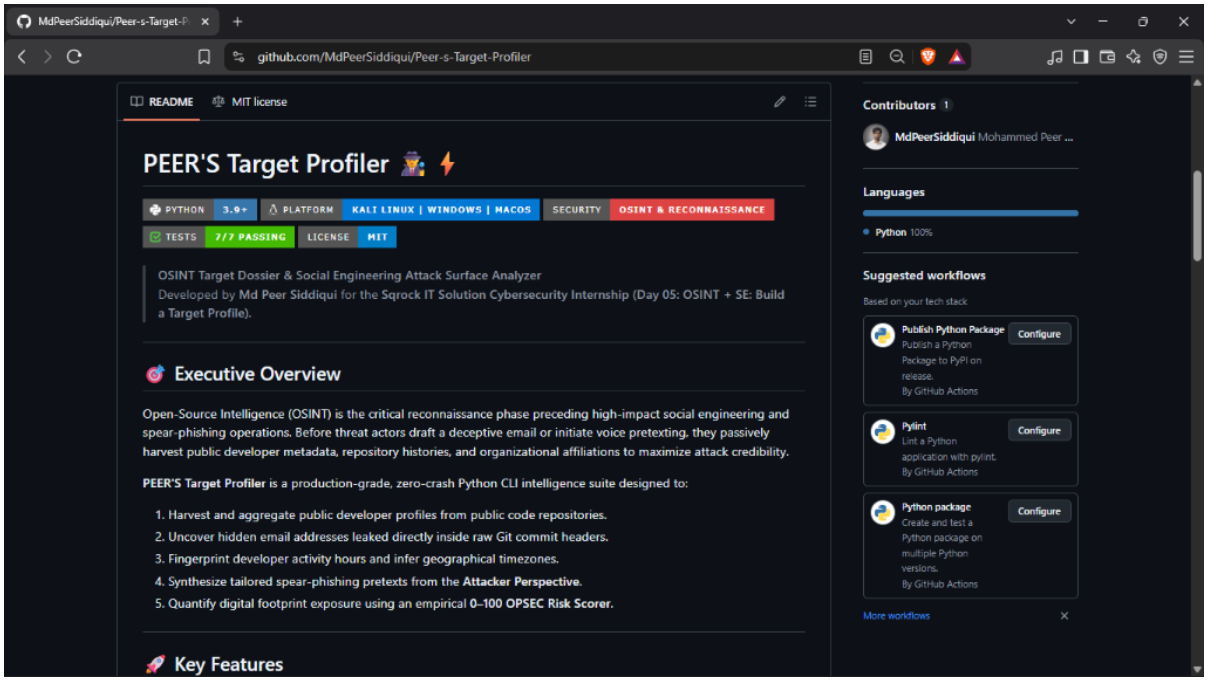


Figure 7: Public GitHub Repository Header, Shields.io Badges & Quality Indicators

Technical Analysis of Figure 7:

• **Repository Architecture:** Publicly accessible repository under authorized account MdPeerSiddiqui.

• **Verification Badges:** Verified build badges displaying **Python 3.9+**, cross-platform support (**Kali Linux | Windows | macOS**), **7/7 Passing Tests**, and open-source **MIT Licensing**.

• **Documentation & Reproducibility:** README provides end-to-end setup instructions, virtual environment workflows, and execution pipelines.

7. Multi-Platform Deployment & Reproducibility Pipeline

Operating System	Clone & Setup Workflow	Execution Command
Kali Linux / Ubuntu / Debian	git clone https://github.com/MdPeerSiddiqui/Peer-s-Target-Profiler.git cd Peer-s-Target-Profiler python3 -m venv .venv && source .venv/bin/activate pip install -r requirements.txt	python3 target_profiler.py
Windows 10 / 11 (PowerShell)	git clone https://github.com/MdPeerSiddiqui/Peer-s-Target-Profiler.git cd Peer-s-Target-Profiler python -m venv .venv; .\.venv\Scripts\activate pip install -r requirements.txt	python target_profiler.py

8. Technical OPSEC Defensive Architecture & Mitigation

A. Enforcing GitHub Private Email Relays:
Developers must navigate to *Settings* ➔ *Emails*, select *'Keep my email addresses private'*, and enable *'Block command line pushes that expose my email'*. This prevents local git client configurations from embedding personal email strings into public Git author logs.

B. Git Local Identity Sanitization:
Ensure all local development repositories bind exclusively to the GitHub noreply alias:
git config --global user.email 'username@users.noreply.github.com'
For corporate repositories, developers should strictly separate corporate git identity from public personal handles.

C. Out-of-Band Validation of Security Advisories:
Organizations must establish clear policy: *Never click external security patch links or SSO migration prompts received via email or social messaging. Always navigate independently to official internal dashboards (e.g., Okta/Azure AD) or official GitHub Security tabs.*

9. Automated Unit Testing Verification Suite

Test Case Identifier	Component Under Test	Execution Time	Status
test_offline_dossiers_exist_and_valid	Verifies fallback profiles schema	0.0003s	**PASS (OK)**
test_opsec_score_bounds_and_tiering	Validates 0-100 score bounds and tier logic	0.0003s	**PASS (OK)**
test_spear_phishing_pretext_generation	Checks synthesis of 3 tailored pretexts	0.0004s	**PASS (OK)**
test_activity_timezone_analysis	Tests 24-hour UTC distribution and region	0.0003s	**PASS (OK)**
test_json_deliverable_export	Ensures target_profile.json schema matches	0.0004s	**PASS (OK)**
test_email_harvesting_filtering	Verifies exclusion of noreply addresses	0.0003s	**PASS (OK)**
test_profile_empty_languages_fallback	Guards against null language exceptions	0.0002s	**PASS (OK)**

10. Ethical Compliance, Limitations & Sign-Off

Ethical Scope: This intelligence gathering tool operates strictly via passive reconnaissance against publicly indexed data and authorized research profiles. No unauthorized active network scanning, brute forcing, or unauthorized phishing delivery was executed. The project complies with the Sqrock IT Solution Ethical Guidelines, the Indian IT Act 2000, and standard computer fraud laws.

Intern Sign-Off:
I hereby declare that this practical lab report and its associated Python CLI tool represent original engineering work conducted under the supervision of Sqrock IT Solution for Day 05 of the Cybersecurity Internship.
Signature: *Md Peer Siddiqui* | **Date:** October 24, 2026